Developer Brief — AI Conversation Monitoring & Color-Coding System

Prepared for: A7 Group Real Estate — AI / Full-Stack Developer
Owner: Malik (Managing Director)
Systems in use: Bitrix24 On-Premise (AWS Lightsail) · 3CX AI Enterprise · WhatsApp Business API · WhatsApp agent numbers linked into Bitrix24 · n8n

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1. Purpose (plain English)

We have ~30 telesales agents (Lahore) and a Dubai closing team. Every day they have hundreds of conversations with clients — over WhatsApp and over phone calls.

Malik **cannot** read thousands of chats or listen to thousands of recordings. So we need an **AI layer** that reads every conversation, understands it, and marks it with a **colour** so Malik can look at one dashboard and instantly see:

- Which conversations are going well (deal closing).
- Which agents are failing to satisfy the client.
- Which clients are being abusive.
- **Which agents are trying to do corruption / close deals outside the office.**

This applies to **both channels**: WhatsApp chats **and** 3CX call recordings (transcripts already exist in 3CX).

The AI must **understand context**, not just match words. Keywords (Section 7) are only helpers — the final colour is decided by reading the whole conversation.

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2. The 5 Colour Categories

Colour	Meaning	When to assign
WHITE	Hot / healthy deal	Client likes the property, both sides professional, closing is being discussed (contract, booking, cheque, viewing confirmed, "I'll take it").

Colour	Meaning	When to assign
GREEN	Normal & clean	Ongoing conversation, no problems, but not yet at closing stage. Nothing to worry about.
YELLOW	Agent not satisfying client	Client asks for something the agent cannot provide, property doesn't match the client's need/budget, agent has no suitable listing, client drifting away because agent can't help. **Capability gap.**
PINK	Client unprofessional	The **client** is rude, abusive, threatening, argumentative, or complaining aggressively. (Flag is on the client's behaviour, not the agent.)

Colour	Meaning	When to assign
RED	Corruption / integrity risk	Agent trying to close the deal **outside the office**, agent + client (or agent + outside agent) colluding, "we'll split it between us", moving the client to a personal WhatsApp/number to bypass the company, hiding the deal from the office. **Highest priority — Malik must see these first.**

> **Priority order for review:** RED → PINK → YELLOW → WHITE → GREEN.

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3. Data Sources

A) WhatsApp

- Agent WhatsApp numbers are already linked into Bitrix24 (chats visible in CRM).
- WhatsApp Business API also available.

- We need the **full chat thread** per contact/lead (both sides, in order).

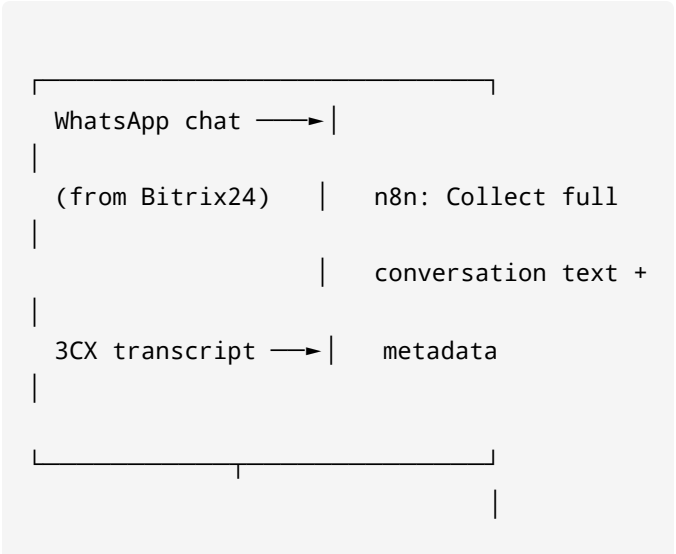
B) 3CX Call Recordings

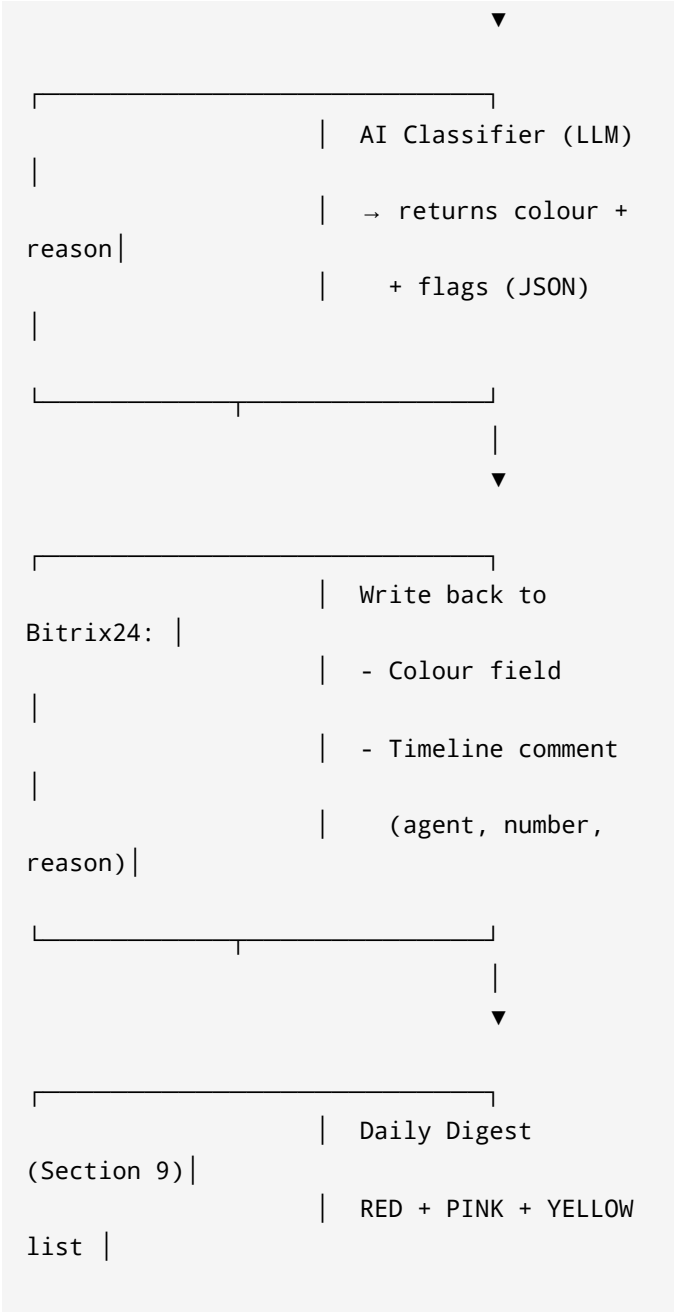
- Recordings **+ AI transcription already produced inside 3CX** — we do **not** need to build speech-to-text.
- We need to pull the **call transcript text +** call metadata (agent name, agent extension, client number, date/time, call ID).

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4. Processing Pipeline (n8n)

Build **one shared classification service** that both channels feed into.




AI Classifier (LLM)
→ returns colour + reason
+ flags (JSON)
Write back to Bitrix24:
- Colour field
- Timeline comment
(agent, number, reason)
Daily Digest (Section 9)
RED + PINK + YELLOW list

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```

Step by step:

1. **Trigger**
    - WhatsApp: on new message / on chat updated for a lead in Bitrix24 (batch every X minutes is fine — no need real-time).
    - 3CX: on new call transcript available (poll 3CX API or export, whichever is accessible).
2. **Collect the full conversation** — not a single message. The AI needs the whole thread / whole call to judge correctly.
3. **Send to AI classifier** with the prompt in Section 6.
4. **Receive structured JSON** (Section 6): colour, confidence, reason, flags.
5. **Write result back to Bitrix24** (Section 8): set the colour field, add a timeline comment with agent name, agent number, client number, the AI's one-line reason, and a short snippet.
6. **Feed the daily digest** (Section 9).

> Keep the classifier as **one reusable n8n sub-workflow / webhook** so WhatsApp and 3CX both call the same brain. Don't build it twice.

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5. Language Handling (important)

- ~95% of conversations are **pure English**.
- ~5% are **Roman Urdu** (Urdu written in English letters, e.g. "aap kaisay ho", "main office main hoon"), sometimes mixed with English.
- **No separate translation step is needed.** A modern LLM (see Section 6) understands English + Roman Urdu + mixed natively.
- Keyword lists in Section 7 are given in **both English and Roman Urdu** to help.

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6. AI Classifier — Prompt & Output

Use a strong general LLM via API (e.g. Claude or GPT-class model). Send the full conversation as input. **Require JSON-only output** so n8n can parse it reliably.

System prompt (give to the model)

```
You are a compliance and quality monitor
for a Dubai real estate agency.
You read one full conversation between a
company AGENT and a CLIENT.
The conversation may be a WhatsApp chat or
a phone-call transcript.
```

```
Language is mostly English, sometimes Roman
Urdu (Urdu in English letters) or a mix —
understand all of them.

Classify the WHOLE conversation into
exactly ONE colour:

- "red"    = Corruption / integrity risk.
The agent is trying to close or move the
             deal OUTSIDE the company:
doing a side deal, splitting money with the
             client or another agent,
taking the client to a personal number to
             bypass the office, hiding the
deal from the company, or colluding.
- "pink"   = The CLIENT is unprofessional:
abusive, threatening, insulting,
             or aggressively
complaining/fighting.
- "yellow" = The AGENT cannot satisfy the
client: no matching property, wrong
             budget/area, client asking for
something the agent can't provide,
             client losing interest because
of a capability gap.
- "white"  = Hot, healthy deal: client is
interested, both sides professional,
             and closing is being discussed
(booking, contract, cheque, deposit,
             viewing confirmed, "I'll take
it").
- "green"  = Normal, clean, ongoing
conversation with no problems and no
closing
```

```
              discussion yet.

Rules:
- Judge by the MEANING of the whole
conversation, not single words.
- RED takes priority over everything. If
there is any genuine sign of a side deal
  or bypassing the office, choose "red".
- Base the decision on what the agent and
client actually mean, even in Roman Urdu.

Respond with ONLY this JSON, nothing else:
{
  "category": "red | pink | yellow | white
| green",
  "confidence": 0.0-1.0,
  "reason": "one short sentence explaining
why",
  "flags":
["side_deal","outside_office","personal_num
ber_shift","money_split","agent_collusion",
"client_abuse","no_matching_property","budg
et_mismatch","closing_signal"],
  "needs_human_review": true or false
}
Include only the flags that actually apply.
Set needs_human_review to true if
confidence is below 0.6 OR the category is
"red".
```

User message (per conversation)

```
CHANNEL: {whatsapp | call}
AGENT: {agent_name} ({agent_number})
CLIENT: {client_number}
DATE: {datetime}

CONVERSATION:
{full chat thread OR full call transcript,
in order}
```

Output → what n8n does with it

- `category` → Bitrix24 colour field.
- `reason` + `flags` → timeline comment.
- `needs_human_review == true` → also push to the daily digest top section, even if not red.

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7. Keyword Reference Lists (helpers, not final decision)

The AI decides by context, but give it — and your own pre-filter/search — these terms in **English + Roman Urdu**. These also help you build a fast keyword search in Bitrix24 later.

RED — Corruption / side deal

- English: outside the office, off the books, between us, keep it between us, we split, share the

commission, do it directly, skip the office, don't tell the office, cash deal, under the table, my personal number, message me on my number, save this number, let's deal privately, without the company, bypass, we'll handle it ourselves

- Roman Urdu: aapas mein, hum dono mil ke, office ko mat batao, office ke bahar, personal number pe baat karo, mera personal number, ye number save karlo, cash mein karte hain, commission aadha aadha, hum le lenge, andar khaate, seedha karte hain, company ko chhod ke

RED — Personal-number / channel shift (bypass)

- English: text me here instead, move to this number, don't use the company number, my WhatsApp is, call me on my mobile
- Roman Urdu: iss number pe message karo, company wale number pe nahi, mere WhatsApp pe, mere mobile pe call karo

PINK — Client unprofessional

- English: worst service, useless, you people are frauds, cheater, I'll report you, scam, waste of time, shut up, don't call me again (aggressive), threats, insults/abuse

- Roman Urdu: bakwaas, faltu, dhoka, fraud ho tum log, tang mat karo, phone mat karo (aggressive), gaali (any abuse)

YELLOW — Agent can't satisfy client

- English: do you have anything else, this doesn't match, not what I asked, out of my budget, too expensive, wrong area, nothing suitable, I wanted something bigger/cheaper, any other option, this isn't for me
- Roman Urdu: koi aur option, ye theek nahi, budget se zyada, mehnga hai, dusri jagah chahiye, kuch aur dikhao, ye wala nahi

WHITE — Hot / closing

- English: I'll take it, let's proceed, send the contract, booking, deposit, cheque ready, when can we sign, ready to move forward, confirm the viewing, let's close
- Roman Urdu: ye pasand aa gaya, aage barhte hain, contract bhejo, booking karte hain, cheque ready hai, kab sign karein, deal pakki

GREEN — Normal

- No flags from above; general Q&A, sending listings, scheduling, follow-up.

> **Note:** A keyword alone must NOT auto-set red. The AI must confirm the *meaning* (e.g. "personal number" said innocently vs. to bypass the office). Keywords just raise a flag for the AI to check.

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8. How It Shows in Bitrix24

1. **Custom field** on the Lead / Deal (and/or Contact) called e.g. `Conversation Health` — a dropdown/enumeration:
   `RED · PINK · YELLOW · WHITE · GREEN`
   Set the enumeration colours to match (or use an emoji dot 🔴🩷🟡⚪🟢 if colour styling is limited).
2. **Timeline comment** added automatically each time:

   ```
   [AI Monitor] {COLOUR}
   Agent: {agent_name} ({agent_number})
   Client: {client_number}
   Channel: {whatsapp/call}
   Reason: {AI reason}
   Flags: {flags}
   ```

3. **Kanban / List view** filtered by `Conversation Health` so Malik opens ONE view and sees all RED at the top, then PINK, then YELLOW.

4. Keep the colour field **updatable** — if a green chat later turns red, the next run overwrites it (keep the latest classification, but the timeline keeps the history).

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9. Daily Digest (the key output for Malik)

Once a day (e.g. 9:00 AM Dubai time), n8n sends Malik **one summary** (WhatsApp to Malik, and/or email, and/or a Bitrix24 task):

```
Daily Conversation Report — {date}

🔴 RED (corruption / side deals): {count}
   1. Agent {name} ({number}) ↔ Client {number} — {reason} — [open in CRM]
   2. ...

🩷 PINK (client abusive): {count}
   1. ...

🟡 YELLOW (agent couldn't satisfy client): {count}
   1. ...

⚪ WHITE (hot / closing): {count}   — good news, top deals to push
🟢 GREEN (normal): {count}
```

```
Needs human review: {count}
```

This is the whole point: Malik reads **one message** and knows "today there are 3 red, 5 hot deals" — instead of thousands of chats and recordings.

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10. Edge Cases & Notes

- **Full thread, not single message.** Always classify on the whole conversation/call, or the AI will misjudge.
- **Confidence threshold.** If confidence < 0.6, set `needs_human_review = true` so Malik can spot-check. All RED always goes to review regardless of confidence.
- **Roman Urdu + English mix** — handled natively by the LLM; no translation step.
- **Cost control.** No need to re-classify a conversation that hasn't changed. Store a hash of the conversation; only re-run when new messages/calls arrive.
- **Call vs chat length.** Very long call transcripts: send the full transcript if within model limit; if too long, send the full transcript in chunks and take the most severe category (RED wins).

- **Employee policy.** This monitoring should be covered by the existing A7 Group team communication-monitoring policy — make sure the field/notes align with what agents were told, since Malik already has that policy in place.

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11. Suggested Build Order (phases)

Phase 1 — WhatsApp only, manual trigger.
Pull one lead's WhatsApp thread → AI classify → write colour + comment to Bitrix24. Prove it works on 10–20 real chats.

Phase 2 — Automate WhatsApp.
n8n trigger on new/updated chats, batched. Colour field live on Kanban.

Phase 3 — Add 3CX calls.
Pull 3CX transcripts into the **same** classifier service. Same colours, same field.

Phase 4 — Daily Digest.
Automated 9 AM report to Malik with RED/PINK/YELLOW lists + CRM links.

Phase 5 — Tuning.
Review a week of results with Malik, adjust the prompt/keywords for false positives (especially RED), lower review load.

12. What We Need From Malik / Access Required

- Bitrix24 API access (webhook) with rights to read chats and write fields + timeline comments.
- 3CX API / export access to pull call transcripts + metadata.
- LLM API key (Claude or GPT-class).
- n8n instance (already running).
- Confirmation of which entity the colour field sits on (Lead vs Deal vs Contact).

End of brief. Questions on any section can go straight back to Malik.